

Captured by the Words: Literal Attraction in Typed Decision Models Tracks the Test Language Rather Than Capability on Bangla Idioms

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Abstract

Decision models, non-generative scorers that report a probability over caller-supplied options and emit no text, are increasingly deployed as routers, classifiers and guardrails. Because they generate nothing, rationales and chain-of-thought do not apply, and idioms make token attribution hard to interpret: for an expression whose figurative reading is not recoverable from its parts, it is unclear what a per-word contribution would mean. We therefore audit three such models, an encoder scorer, a decoder scorer and a closed commercial API, entirely through interventions on the option set, using expert-annotated Bangla idioms that ship with both a gold figurative gloss and a gold literal gloss. Dropping an idiom’s own literal gloss into the option list captures all three models. Running the full model $\times$ language matrix over one shared item set shows the failure tracks the test language rather than capability: every arm is several times more captured in Bangla than in English on the same paired entries, and the commercial model’s apparent resistance turns out to be a property of the language it was tested in. Confidence inverts under the same shift, so gating on it does worse than gating at random, and on a semantics-free option list, where no option carries information and expected accuracy is 0.25, the best-calibrated arm still reports 0.82 confidence. Option-order sensitivity, by contrast, separates the two scoring architectures rather than the three models.

1 Introduction

A *decision model* is given a question and a list of candidate options and returns a probability distribution over exactly those options. It is not autoregressive, it does not sample, and it produces no text. This class has grown quickly as infrastructure for routing, intent classification, content gating and tool selection, precisely because it is cheap,

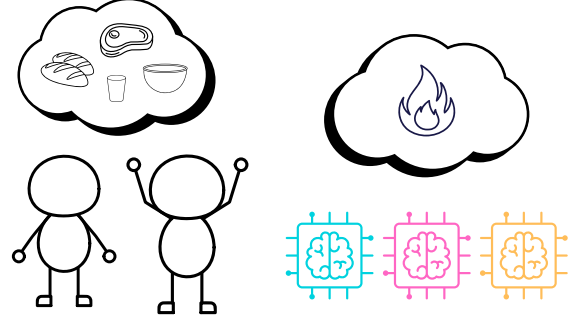


Figure 1: **The model takes the words.** *Laya-ml*, *Lod-lille*, and *Jev-1.13* are shown in cyan, magenta, and orange, respectively, throughout the paper. The Bangla idiom *অগ্নিবল* (Agnibal) is composed of *অগ্নি* (*fire*) and *বল* (*strength*), but means intense hunger; English expresses the same concept as *fire in the belly*. When both readings are placed in the option list, all three models choose the literal gloss in Bangla, at $p = 1.00, 0.97$, and 0.36 , respectively; with the same four options in English, two of the three recover. Across 3,265 paired items, literal attraction falls from 0.731, 0.918, and 0.304 in Bangla to 0.379, 0.241, and 0.047 in English (§5.1).

bounded and returns a number a caller can threshold on. An ecosystem analysis of 2,170 public projects finds it already embedded as a reusable decision component across domains (Ling et al., 2026), and proposals now place these models at the adjudication points of autonomous security harnesses (dos Santos Barbosa, 2026). The threshold is load-bearing.

That bounded interface is also an explainability problem. There is no rationale to inspect, no chain of thought to measure for faithfulness (Jacovi and Goldberg, 2020; Turpin et al., 2023), and in the commercial case no weights at all. What is left is the behavioural surface: the caller controls the question and the option list, and sees a probability vector. Any audit of this class has to be built out of that.

The obvious fallback, attributing the decision to input tokens (Ribeiro et al., 2016; Lundberg and Lee, 2017), is both unreliable here (Jain and Wal-

lace, 2019; Adebayo et al., 2018; Hooker et al., 2019) and hard to read. How far an idiom’s parts map onto components of its figurative meaning varies by idiom and is not annotated in our corpus (Nunberg et al., 1994; Gibbs et al., 1989); for an expression at the non-decomposable end, a Shapley value over its words is computable but has no evident linguistic referent. We would rather intervene on the expression as a unit than report shares we cannot interpret (Appendix D).

We take the option set as the intervention surface instead. Bangla *bāgdhārā* (idioms) are well suited to this because the expert resource we use ships, for each idiom, both a gold figurative meaning and a gold *literal* gloss: a ready-made, human-certified counterfactual distractor that no annotation effort of ours produced. Dropping that gloss into the option list turns a benchmark question into a targeted probe: does the model read the expression as an idiom, or as its words? Figure 1 shows one item answered both ways.

We contribute (i) a meaning-choice instrument for decision models in which the *distractor family* is the independent variable and everything else is held fixed, requiring no new annotation (§4); (ii) the Literal Attraction Rate, measured as a full model $\times$ language matrix, which shows capture tracking the test language rather than capability (§5.1); (iii) evidence that confidence is anti-diagnostic under this shift across three different calibration philosophies, with confidence-gated selective prediction worse than random gating (§5.2); (iv) a large dissociation in option-order dependence between the two scoring designs, with a methodological consequence for anyone evaluating this class (§5.4); and (v) a semantics-free control, the one condition that runs identically on all three arms, bounding what each model reports when the options carry no information at all (§5.3).

2 Related Work

The decision-model class. The evaluation literature is still forming, and reports competitive quality at far lower cost with no independent accuracy advantage from the typed readout itself (Tang and Zheng, 2026; Rao and Callison-Burch, 2026; Ibrahim and Zaki, 2026; Guo et al., 2026; Yu and Yao, 2026). Two strands bear on ours. Ibrahim and Zaki (2026) report decision-model *calibration exceeding most LLM baselines*, and Li et al. (2026) build the confidence-gated cascade that follows

from it; both depend on the returned confidence being diagnostic, and §5.2–§5.3 give conditions under which it is not. Concurrently Sun et al. (2026) find the decision follows an option’s *name* rather than its bound definition, and Xu (2026) flip 61.4% of correct decisions with short, fluent context additions chosen to redirect them. Appendix B sets out how each relates to our design.

Idioms in NLP. Idiom resources and tasks are well established (Haagsma et al., 2020; Tayyar Madabushi et al., 2022; Chakrabarty et al., 2022), including for Bangla (Sakhawat et al., 2026), and the difficulty of figurative language for distributional models is long-noted (Shwartz and Dagan, 2019). Psycholinguistics frames the phenomenon as competition between a literal and a figurative route whose resolution depends on decomposability and context (Cacciari and Tabossi, 1988; Gibbs et al., 1989; Titone and Connine, 1999), and Oh et al. (2026) give mechanistic evidence that the competition is real inside a transformer: causal tracing recovers two pathways carrying the figurative and literal readings simultaneously, with early layers promoting the figurative one and a direct route keeping the literal one available.

Our D-LIT intervention is the behavioural counterpart of that picture rather than a replication of it: they can ask *how* the literal route stays available inside a model they can open, we can only ask *how often* it wins in one that emits no text and, in one arm, exposes no weights (Appendix B).

Evaluating option scorers. Option-order sensitivity and selection bias are documented for prompted generative LMs (Pezeshkpour and Hruschka, 2024; Wei et al., 2024; Zheng et al., 2024; Zhao et al., 2021). We are not aware of a comparison of option-marker against independent-option scoring in small non-generative scorers over the full permutation set, which is what §5.4 provides.

Calibration and selective prediction. Proper scoring and calibration measurement are standard (Brier, 1950; Gneiting and Raftery, 2007; Guo et al., 2017; Nixon et al., 2019; Kumar et al., 2019), as is the observation that calibration degrades under distribution shift (Ovadia et al., 2019; Jiang et al., 2021). Selective prediction gives the operational test (Geifman and El-Yaniv, 2017, 2019; Traub et al., 2024). We report the combination that matters for deployment: accuracy *decreasing* with confidence, and a confidence gate underperform-

ing a random one.

Behavioural testing. Our design follows behavioural checklists, contrast sets and counterfactual minimal pairs (Ribeiro et al., 2020; Gardner et al., 2020; Kaushik et al., 2020; Naik et al., 2018; Jia and Liang, 2017); the contribution is carrying that tradition into a model class with no text output, where it is one of the few instruments left.

Low-resource evaluation. Bangla has strong encoders and corpora (Bhattacharjee et al., 2022; Kakwani et al., 2020), and cross-lingual benchmarks establish the transfer gap (Hu et al., 2020; Ruder et al., 2021; Adelani et al., 2021); §5.7 localises our failure to the Bangla side rather than to figurativeness.

3 Dataset

Source corpus. We use *Bangla Bagdhara* (Sakhawat et al., 2026), a corpus of 10,361 Bangla (Bengali) idioms annotated under a 19-field schema by expert consensus.¹ Each row carries the idiom, a gold figurative meaning in Bangla, an expert literal gloss, semantic tags, alternative surface forms certified as meaning-preserving, and a gold carrier sentence. We add no annotation; every intervention below is assembled from fields that already exist.

Its own benchmark on the same underlying task reports no model among 30 multilingual and instruction-tuned LLMs above 50% accuracy, against 83.4% for human annotators. Option construction differs, so those figures are not directly comparable to ours, but they locate the difficulty in the material rather than in our instrument.

4 Methodology

Figure 2 sets out the instrument end to end: what the corpus supplies, what each condition changes, and what comes back.

4.1 The meaning-choice task

Given an idiom, choose its gold figurative meaning from $k = 4$ candidates. Chance is 0.250 throughout. The **distractor family is the independent variable**; the item, the instruction, k , and per-item option-order randomisation (seeded by item id) are held fixed across families, so a paired test across families is a test on byte-identical questions.

¹Two rows are lost to parsing, leaving 10,359; of these 8,876 carry every field the instrument requires.

4.2 Distractor families and measures

The five families are D-RAND (other idioms’ glosses at random), D-SURF (glosses sharing surface tokens with the gold), D-SEM (glosses from idioms sharing semantic tags), D-LIT (one distractor replaced by the idiom’s *own* expert literal gloss) and D-HARD (all three pressures at once). Appendix C gives each sampling rule, and Table 7 one complete item in the original script.

For the families containing a literal gloss, with ℓ_i the literal option and g_i the gold, the Literal Attraction Rate is

$$\text{LAR} = \frac{1}{N} \sum_{i=1}^N \mathbf{1} \left[\arg \max_o p_i(o) = \ell_i \right].$$

LAR is not $1 - \text{accuracy}$: it isolates the errors that are specifically a literal reading, and has no floor, since a model that never reads literally scores 0 while still being wrong.

4.3 Interventions and statistics

Seven interventions vary one thing each: the distractor family (E1), the idiom’s surface form (E2), the idiom as a unit (E3), the task language (E4), nothing (E5, derived from E1), the option text (E6) and the option order (E7). Appendices C and A give what each holds fixed and the typed question they share.

All intervals are 95% percentile bootstrap over items (2,000 resamples); paired comparisons use McNemar’s exact test; ECE uses 15 equal-width bins and AURC is reported against a random-gate baseline. The two open arms were built independently from a written contract, and on all 4,440 overlapping rows the gold key, gold position, idiom string and literal-distractor key agree at 1.000, so every paired test is on identical questions (Appendix R).

4.4 Models

Three models span the design space (Table 1). **Laya-ml** is a bidirectional encoder scoring all options together from marker positions. **Lod-lille** is a decoder with independent-option scoring, each option attending only to the shared state, the question and itself from the same start position, which makes it order-independent by construction in fp32. **Jev-1.13** is a closed commercial model behind a decisions API, whose documentation names English as its primary and best-supported language; we know its interface and

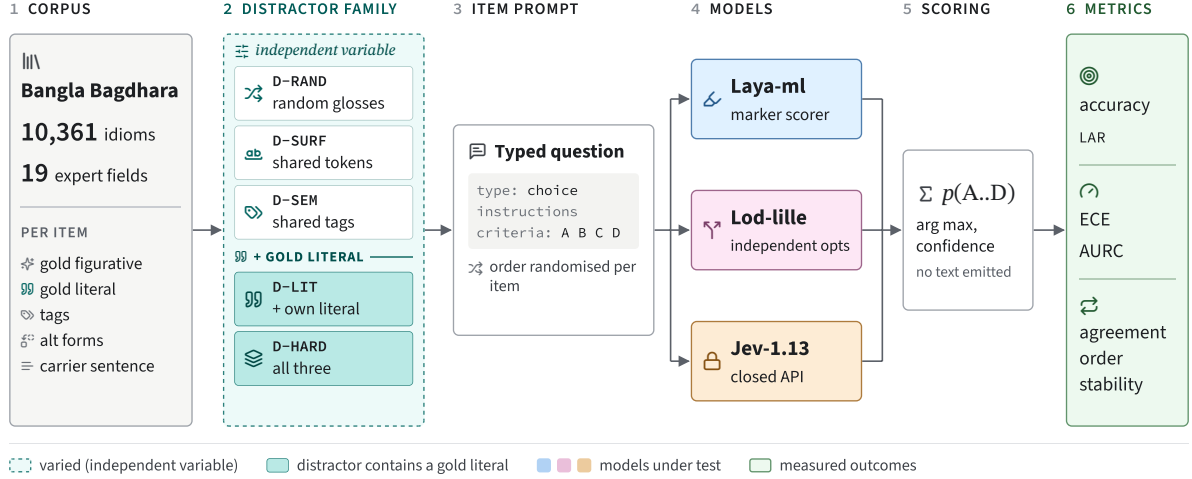


Figure 2: **The instrument in one view.** The audit proceeds from the **Bangla Bagdhara** corpus through six stages: **(1) corpus**, which supplies 10,361 expert-annotated idioms with gold figurative and literal glosses; **(2) distractor family**, the sole manipulated variable, with D-LIT and D-HARD injecting the expert literal gloss; **(3) item prompt**, which fixes the typed question, $k = 4$ options, instructions, and per-item order seed; **(4) models**, comprising Laya-ml, Lod-lille, and Jev-1.13; **(5) scoring**, which records the four-option probability vector, arg max, and confidence without requiring generated text; and **(6) metrics**, which evaluate accuracy, LAR, ECE, AURC, agreement, and order stability. The figure’s visual encoding distinguishes the varied distractor family, literal-bearing conditions, models under test, and measured outcomes.

	Laya-ml	Lod-lille	Jev-1.13
backbone	mmBERT	Qwen3-0.6B	closed
scoring	marker	independent	unknown
parameters	322M	0.6B	n/a
items	8,876	888	8,761
latency	27.2 ms	86.1 ms	API
weights	open	open	closed

Table 1: **The three model arms.** Laya-ml is a 322M-parameter mmBERT marker scorer, Lod-lille is a 0.6B-parameter Qwen3-based independent-option scorer, and Jev-1.13 is a closed API with an unknown scoring architecture. The item counts and latencies are those used in the reported experiments. Lod-lille was run on a fixed 10% random subset; Appendix O bounds the resulting drift at 0.016.

Distractors	Laya	Lod	Δ
random	0.367	0.484	+0.133***
surface	0.338	0.412	+0.073***
tag	0.333	0.446	+0.120***
+ literal	0.151	0.066	-0.070***
hard	0.144	0.063	-0.086***
LAR (lit.)	0.644	0.885	
LAR (hard)	0.631	0.855	
mean p (lit.)	0.464	0.726	
mean p (gold)	0.195	0.128	

Table 2: **The distractor ladder and literal attraction.** McNemar tests use the 888 shared items (***) $p < 0.001$). Cyan marks Laya-ml, while magenta marks Lod-lille; darker cyan and magenta fills highlight the higher value within a row. The highlighted literal-bearing conditions and capture measures expose the central pattern: adding the idiom’s own literal gloss causes a sharp accuracy drop and strong literal attraction. The sign of Δ reverses at the literal boundary.

nothing else. Appendix A gives the call each one receives.

5 Analysis

5.1 Literal capture, and what it tracks

Table 2 and Figure 3 give the ladder for the two open arms on the 888 shared items. Lod-lille is the stronger model on every non-literal condition, by up to +0.133. Insert the idiom’s own literal gloss and the ordering *reverses*: Lod-lille falls to 0.066, less than half of Laya-ml’s already-below-chance 0.151, and barely a quarter of chance. The same reversal holds for D-HARD.

Within this pair, then, LAR measures something accuracy does not: a model can be selected as

“better” on a conventional benchmark while being worse at what that benchmark proxies for. Laya-ml and Lod-lille answer byte-identical questions in the same language, so the crossover between them is a paired result. The error is not diffuse: the decomposition under D-LIT is gold / literal / other = 0.151 / 0.644 / 0.207 for Laya-ml and 0.066 / 0.885 / 0.048 for Lod-lille, and on the clearest items Laya-ml places $p = 1.000$ on the literal gloss.

Capture tracks the language, not capability. The arms above ran the literal condition in Bangla;

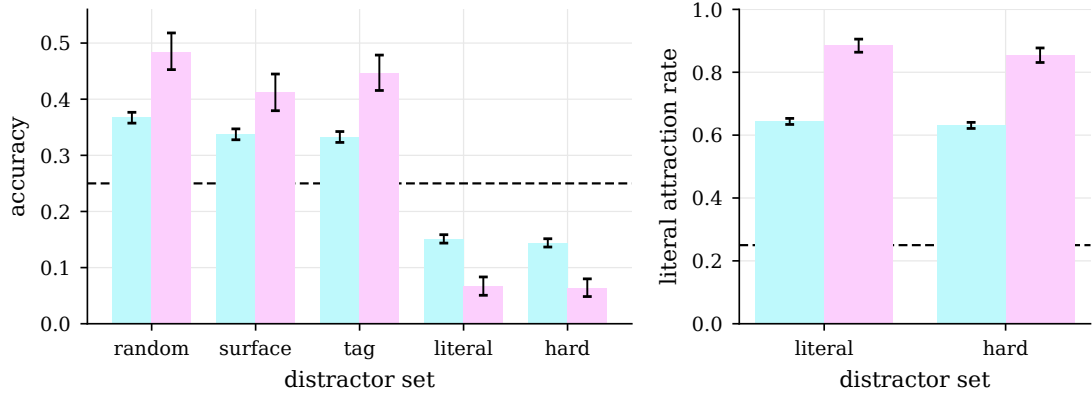


Figure 3: The distractor ladder (left) and literal attraction rate (right) for the two open arms. ■ Laya-ml ($n = 8,876$), ■ Lod-lille ($n = 888$); the black dashed horizontal line marks the four-option chance level (0.250) in both panels; error bars are 95% bootstrap CIs. Surface- and tag-matched distractors cost a few points. Inserting the idiom’s own literal gloss (D-LIT) collapses both models far below chance, and collapses the *stronger* model harder: Lod-lille leads Laya-ml by +0.133 on random distractors and trails it by -0.070 once the literal gloss is present. On the right, both models select the literal reading over the figurative one on the majority of items.

the commercial arm’s headline $LAR = 0.047$ is *English*, so the two are not comparable as they stand. We therefore ran the full model $\times$ language matrix on the 3,265-item shared pool, every cell built by one function from one seed (Table 15, Figure 4). The English cell for Jev-1.13 reproduces its published value at 0.0472 [0.0401, 0.0545], which is the check that the two constructions measure the same thing.

Two things follow. First, **every arm is far more captured in Bangla than in English on identical items**: $0.731 \rightarrow 0.379$, $0.918 \rightarrow 0.241$ and $0.304 \rightarrow 0.047$, all paired McNemar $p < 10^{-100}$. The commercial model’s apparent resistance to capture was a property of the language it was tested in. Second, **capture is not monotone in capability**: Lod-lille is more accurate than Laya-ml and more captured, while Jev-1.13 is more accurate than both and less captured, so capability neither predicts capture nor protects against it. The two open arms fall below chance here; Jev-1.13 does not, but loses 0.390 of accuracy against its own English run on the same items. Appendix I gives the cells with intervals, and the caveat that the pool is not a neutral sample.

The capability ordering. One condition is identical across all three arms, Bangla idiom with random distractors, and on it the commercial model is far ahead, 0.736 against 0.367 and 0.484 (Table 3). That is the ordering against which the calibration results below should be read, and it is the one capture does not follow.

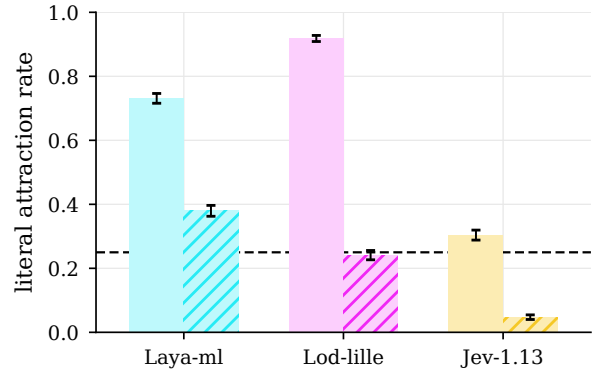


Figure 4: Literal attraction by model and task language, all six cells measured on the same 3,265 items ($n = 3,265$ each; error bars are 95% bootstrap CIs). The black dashed horizontal line marks the four-option chance level (0.250). ■ Laya-ml, ■ Lod-lille, and ■ Jev-1.13; solid bars are the Bangla condition, while hatched bars are the English condition. Every arm is far more captured in Bangla than in English on identical items, and the Bangla ordering does not follow capability: Jev-1.13 is the most accurate of the three and the least captured, while Lod-lille is more accurate than Laya-ml and the most captured. Jev-1.13’s English value reproduces the figure reported from its original run, 0.047. Full numbers are given in Table 15.

5.2 Confidence is anti-diagnostic

Table 4 gives the mean probability assigned to the chosen option, split by outcome, on the literal-bearing families. Both open models are *more confident when captured than when correct*, Lod-lille extremely so (0.787 against 0.453). The reliability curves in Figure 5 slope *downward* under D-LIT in both.

Selective prediction therefore inverts: restricting Laya-ml’s D-LIT to its most confident 5% gives 0.007 accuracy against 0.151 at full coverage, and

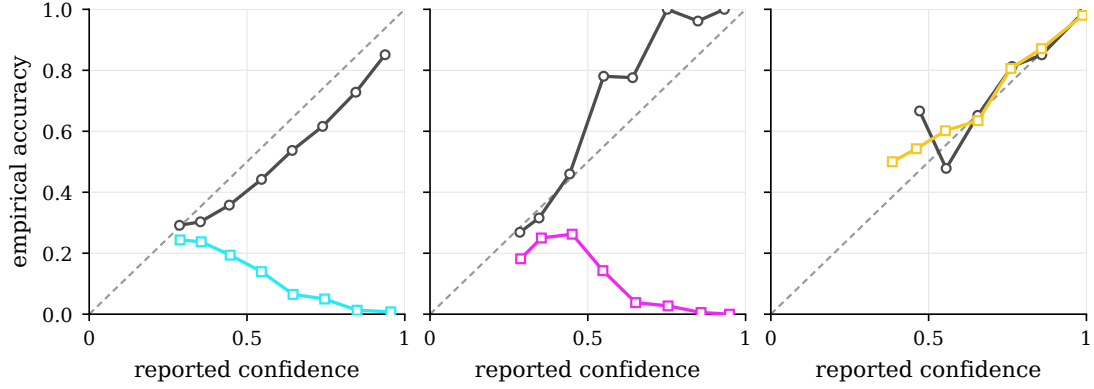


Figure 5: Reliability, 15 equal-width bins. Panels left to right: Laya-ml, Lod-lille, Jev-1.13. ■ circles are the random-distractor condition; squares in each panel’s own colour are the +literal condition. The grey dashed diagonal is perfect calibration. Under D-LIT the two open models’ curves bend *downward*, higher stated confidence buying lower accuracy, while Jev-1.13, running in English, stays on the diagonal. Reading the left two panels: the model is most wrong exactly where it reports being most sure.

Model	n	Accuracy	95% CI
Laya-ml	8,876	0.367	[0.357, 0.377]
Lod-lille	888	0.484	[0.453, 0.518]
Jev-1.13	8,761	0.736	[0.727, 0.745]

Table 3: The only condition identical across all three arms: Bangla idiom $\rightarrow$ Bangla gloss, random distractors. All pairwise McNemar tests on the overlapping items are significant: Jev–Laya +0.369, Jev–Lod +0.259, Lod–Laya +0.133 (all $p < 10^{-9}$).

Outcome	Laya	Lod	Jev
correct	0.432	0.453	0.935
captured	0.596	0.787	0.737
other error	0.421	0.399	0.746
ECE (D-RAND)	0.071	0.073	0.005
ECE (D-LIT)	0.385	0.679	0.014
AURC (conf.)	0.924	0.989	0.013
AURC (random)	0.849	0.933	0.069

Table 4: Confidence by outcome on the literal-bearing families, and calibration. Bold marks, per model, whether the model is more confident when correct or when captured. Both open arms are more confident when captured; their confidence-gated AURC is *worse* than the random-gate baseline in the grey row, which inverts selective prediction. Jev-1.13’s English condition does not show the inversion.

Lod-lille answers every item in its most confident 35% incorrectly (Appendix F). This is not an artifact of one calibration philosophy. Laya-ml ships fitted per-bucket temperatures, Lod-lille ships none by design, and Lod-lille’s *separate confidence head*, meant to estimate whether the answer is in the option list at all, shows the same inversion (0.631 on captured items against 0.295 on correct ones).

Jev-1.13 is the interesting exception, and it is only an exception in one direction. Its reliability

in English is close to ideal on the two families in Table 4 (ECE 0.005 and 0.014, rising to 0.047 on D-HARD), and its confidence-gated AURC beats random by an order of magnitude. That makes §5.3 harder to dismiss, not easier.

This does not contradict Ibrahim and Zaki (2026), who find decision-model calibration better than most LLM baselines; we reproduce that for the commercial arm on its own language. The narrower claim is that in-distribution calibration need not survive a targeted distractor shift, and that it fails in the costly direction: a confidence-gated cascade (Li et al., 2026) accepts the high-confidence tail without review, and on the literal-bearing families that tail is where the open arms are *least* accurate.

5.3 A semantics-free control, on all three arms

The o4 ablation replaces every option’s text with a bare label. No semantic signal remains, so with the gold position randomised the expected accuracy is 0.25 absent any signal, and whatever confidence survives is overconfidence with no modelling assumption needed to read it. This is the only condition in the paper that runs identically on all three arms: it requires no gloss, no literal annotation and no shared language, so the language confound that scopes §5.1–§5.2 to the two open arms does not arise here. Figure 6 reports it.

The ordering reverses against Table 4. The best-calibrated model under real options is the worst when the options carry nothing: Jev-1.13 reports 0.815 mean confidence at 0.254 accuracy, an ECE of 0.561. Lod-lille, the weakest of the three on accuracy, is the only one whose reported confidence

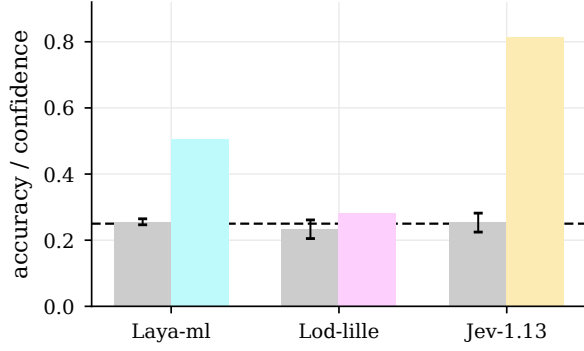


Figure 6: The semantics-free control. Per model, the grey bar is accuracy (with 95% bootstrap CI) and the coloured bar is mean reported confidence (cyan Laya-ml, magenta Lod-lille, yellow Jev-1.13). The black dashed horizontal line marks the four-option chance level (0.250). Because the options carry no information, accuracy is at chance by construction and the gap between confidence and accuracy is pure overconfidence: 0.250 for Laya-ml, 0.049 for Lod-lille, and 0.561 for Jev-1.13, which reports 0.815 confidence while answering at 0.254. Jev-1.13’s condition is English.

collapses appropriately (0.282). We read this as the most portable finding in the paper, because it is the one that needs no annotation and no matched language to reproduce: a calibration number measured on in-distribution options says little about what a model will report when the option list is degenerate, and a caller thresholding on a returned probability is exposed to that case. Confidence-based acceptance is the mechanism of the cascade designs proposed for this class (Li et al., 2026; Ibrahim and Zaki, 2026), and a degenerate option list is an ordinary consequence of an upstream bug or an empty retrieval rather than an adversarial construction.

A side-check comes with it: McNemar between the two open arms under o4 is not significant ($p = 0.76$), so the instrument’s null behaves identically in both.

5.4 Option order is architecture-specific

We run every item under all $4! = 24$ orderings of a fixed option set and record the identity of the chosen gloss, recovered as $\text{perm}[\text{slot}(\hat{y})]$.²

Table 5 gives a large dissociation: a $7.8\times$ difference in answer stability and a $12\times$ difference in probability stability. Each scoring design is rep-

²An earlier version of this analysis counted distinct option keys (slot letters) and reported near-total instability for Laya-ml. That metric is an artifact: a key necessarily changes when the same gloss moves slots. All numbers here use the corrected gloss-identity metric; Laya-ml’s figure drops from 1.000 to 0.837 and the finding stands.

	Laya	Lod	Jev
answer changes	0.837	0.107	0.133
correctness varies	0.686	0.070	0.120
distinct glosses	2.83	1.14	1.15
perfectly stable	0.163	0.893	0.867
$p(\text{gold})$ range	0.346	0.029	0.128
selection spread	0.226	0.052	0.072

Table 5: **Option-order sensitivity.** Each item is evaluated under all 24 orderings of the same four options ($n = 992/299/300$ items for Laya, Lod, and Jev, respectively). Cyan Laya, magenta Lod, and yellow Jev identify the three model arms. Bold marks the largest value in each row. The marker-based scorer changes its chosen gloss on 83.7% of items, while the independent-option scorer changes it on 10.7%. Lod is documented as order-independent by construction; the audit in Appendix Q shows that it is not, which is why its entries here are small rather than zero.

resented by exactly one model here, so the design cannot be separated from the backbone or the capability of the model carrying it; we read the pattern as consistent with the scoring architecture rather than as demonstrated to follow from it.

The invariance claim does not hold. Lod-lille’s design is documented as order-independent by construction: each option is scored in its own forward pass, so permuting the list should return identical probabilities. Mapping every slot’s returned probability back to the option that slot displayed, we can ask how far a *single* option’s own score travels across the 24 orderings of its list. At the median item it travels 4.0×10^{-2} , and on none of the 299 items does it stay below 10^{-4} (Appendix Q). That is orders of magnitude above fp32 accumulation noise, so the residual instability in Table 5 is not a tie-breaking or slot-indexing artifact of our harness; the scores themselves move. The reading we take is the narrow one: the independent-option design attenuates order dependence substantially, by roughly an order of magnitude against the marker-based scorer, but it does not remove it, and an evaluation resting on the documented guarantee would be resting on something we could not reproduce.

Positional bias follows. With the gold placed uniformly, Laya-ml takes slot B 37.2% of the time against 14.6% for slot A, a $2.6\times$ accuracy swing from position alone (Appendix Q).

Methodological consequence. Any evaluation of this class that fixes option order is partly measuring position. All results here randomise order per item, seeded by item id, and are averages over

position rather than measurements at one of them; on the evidence here we would treat randomising order as a default for this class rather than as a robustness check.

5.5 Options are read as a bag of words

The o-series ablates the option text while holding the idiom fixed (Figure 10; Table 11 shows the forms applied to real Bangla). Truncating each option to six words and *scrambling the words inside it* cost almost nothing: Laya-ml retains 0.967 and 0.925 of its signal, Lod-lille 0.909 and 0.948, Jev-1.13 is flat to within noise. What all three do on the option side is, behaviourally, close to order-free lexical matching; we observe the invariance and do not claim to have identified the mechanism producing it. The drop to the grey semantics-free bar is the only large one, which certifies the instrument: the task measures option semantics, as a bag.

This converges with Sun et al. (2026), who permute which option name is bound to which definition and find the decision follows the name: within an option, these models read lexical identity rather than composed meaning.

5.6 Neither open model treats the idiom as a unit

E3 perturbs the idiom while holding the option list fixed (Figure 9). This is the experiment that stands in for token attribution: rather than asking what each word contributes, which presumes a decomposition the expression may not support, we ask whether the model behaves as though it stores the expression as a unit. A model that did would lose the figurative reading when the unit is broken.

Neither does. Word-shuffling leaves Laya-ml’s accuracy statistically unchanged (0.158 → 0.168) and reversal helps slightly (0.180); Lod-lille moves within noise. Agreement with the intact answer under shuffling is 0.742 and 0.917, so Lod-lille is more *stable* but no more *compositional*. Table 11 shows the perturbed strings.

The one manipulation that clearly helps both is embedding the idiom in its gold carrier sentence, +0.080 and +0.117: context, not the idiom string, carries the usable signal. The pattern is consistent with these models matching whatever lexical material is available rather than retrieving a stored figurative entry, though E3 can rule out unit storage without establishing what replaces it. That cuts both ways, since Xu (2026) flip 61.4% of initially correct decisions with short, fluent additions cho-

sen to redirect the decision, and nothing in the interface distinguishes a helpful addition from a harmful one.

5.7 The failure localises to Bangla, in every arm

Holding the model fixed and swapping the task language (Figure 7) moves every arm by a large margin: Laya-ml +0.246, Lod-lille +0.380, Jev-1.13 +0.233 (all McNemar $p < 0.001$). As in the matrix, the English condition presents the annotator’s nearest English idiom rather than a translation, so this is a comparison between paired entries; Appendix J sets out what that does and does not license. Lod-lille reaches 0.864 in English against 0.484 in Bangla, so the figurative concepts appear to be largely available to it, and the Bangla route is where the loss occurs. Its documentation states that training evaluation was English-only, which is consistent with what we see from outside the model.

Laya-ml’s control arm supports this reading: its English-only checkpoint run on Bengali script sits at 0.283, barely above chance, which reproduces a documented failure mode and makes it unlikely that the Bangla result is an artifact of the scorer rather than of the language.

The two language cells are paired entries rather than identical items, and their literal glosses are built from one annotated field by different routes; Appendix J documents the asymmetries and shows the capture gap survives on the subset where the largest of them cannot operate.

Jev-1.13 complicates the reading and then confirms it. A model with no multilingual claim outscores both that have one on the Bangla task, so the gap is reachable rather than intrinsic to the script; yet its own literal attraction rises from 0.047 to 0.304 when the same items are asked in Bangla. Being better at a language is not the same as being robust in it, and the penalty reaches every arm we tested.

6 Discussion

The matrix in §5.1 is the result we would defend hardest, because it separates two explanations that are easy to conflate. Read only in English, the commercial arm looks immune to literal capture; read in Bangla on the same items it is captured at six times the rate. What a decision model does with a non-compositional expression is therefore not a fixed property of the model, and a benchmark run

in one language will not tell you how it behaves in another. Capability does not order the failure either: the strongest arm is the least captured, the middle arm the most.

The other results sit at different scopes and are worth keeping apart. The semantics-free control runs identically on all three arms and needs no annotation or shared language, so it is the one we would most expect to transfer. The confidence inversion is a paired result on the two open arms. Option-order dependence tracks the scoring architecture rather than the model, and is the one place our audit confirms a design claim rather than denting one.

For practitioners we would draw three cautions rather than recommendations: a returned probability is worth little without a measurement of what the model returns on a degenerate option list; a benchmark score is not by itself evidence against literal capture, since on our two open arms the two move in opposite directions; and an evaluation that fixes option order is partly reporting position.

This is a boundary condition rather than a rebuttal: the cost and latency advantages are established (Tang and Zheng, 2026; Rao and Callison-Burch, 2026), and what we add is that the property the cascade designs depend on degrades first and silently. The typed interface guarantees a well-formed answer and nothing about whether it was arrived at for a reason, so each of these failures is invisible to a caller who only checks that the output validates.

7 Conclusion

We audited three decision models on Bangla idioms using only interventions on the option set, a method that needs no annotation and stays available when a model emits no text and its weights are closed. Replacing one distractor with an idiom’s own expert literal gloss captures all three, and running the full model $\times$ language matrix on a shared 3,265-item pool shows what the failure tracks: literal attraction falls from 0.731, 0.918 and 0.304 in Bangla to 0.379, 0.241 and 0.047 in English, so the commercial arm’s apparent resistance belongs to the language it was tested in rather than to the model. Capability does not order the failure, since the strongest arm is the least captured and the middle arm the most. Confidence inverts under the same shift, and with the options stripped of semantics, where accuracy is pinned at chance, the arms still report between 0.28 and 0.82 confidence.

Option-order sensitivity separates the two scoring architectures rather than the three models. The instrument costs no annotation and, for the commercial arm, about eleven cents to run; we release it so the same audit can be pointed at other models and other languages.

Limitations

All three models are evaluated zero-shot, and this is the limitation we would weigh most heavily after the one below. Decision models are trained on a particular set of decision tasks, and the open arms’ own documentation reports that base checkpoints are weak on tasks outside that set. Some part of what we measure may therefore be out-of-distribution task transfer rather than literal capture specifically. The o4 control bounds this from one side, since it shows the instrument’s null behaving identically in both open arms, and the D-RAND to D-LIT contrast is within-model and within-item, which removes any fixed task-transfer offset. Neither argument rules the confound out. A single lightly fine-tuned arm would settle it, and we have not run one.

Laya-ml’s run had its per-bucket temperatures unset, so its ECE figures are raw. Lod-lille ships no temperature by design, so both open arms are effectively untempered and comparable on that axis; Jev-1.13’s internal calibration is unknown to us.

Lod-lille ran a fixed 10% subset (888 items), so its intervals are roughly ± 3 points and its stratified cells are thin. Appendix O bounds the subset bias at 0.016 and shows it changes no sign or ordering. E3 on the Lod-lille arm is 180 items and several of its perturbation contrasts are not individually significant. E7 covers 992 / 299 / 300 items rather than the full sets.

The matrix in §5.1 runs on the 3,265 items that carry a usable English literal gloss, which is not a neutral sample: for Laya-ml, in-pool LAR is higher than out-of-pool while D-RAND accuracy is unchanged, so the absolute rates there are higher than they would be corpus wide. The language comparison is paired within item and within model, so that selection affects the levels rather than the direction. The matrix also covers the literal condition only; we did not run the other distractor families in both languages. The confidence results in §5.2 remain two-arm, since the commercial arm’s reliability was measured on its English run.

The two language cells are also not symmetric

in construction, and this is the limitation we would flag hardest. They are paired entries rather than identical items: the Bangla state is the idiom, while the English state is the annotator’s nearest English idiom rather than a translation of it. The two literal glosses are drawn from one annotated field in different ways, and the resulting Bangla gloss shares a content word with its own state on 56.2% of pool items against 25.1% on the English side, which hands a lexical matcher a surface hook in Bangla that it does not have in English. Appendix J sets out all four asymmetries we could find and tests this one directly: restricted to the 1,429 items where the Bangla literal gloss shares no content word with the idiom, the capture gap is +0.358, +0.610 and +0.310 for the three arms, against headline gaps of +0.352, +0.677 and +0.256. The hook therefore does not account for the result, but we cannot rule out smaller construction effects of the same kind, and a properly translation-paired corpus would settle what we can only bound.

More broadly, this is one corpus, one language, two small open models and one commercial endpoint. The semantics-free control is the only result here that we would expect to transfer as stated; the rest should be read as findings about these systems on this material, and as an instrument other people can point at theirs.

Distractors are drawn from the same corpus, so difficulty is corpus-relative rather than absolute. Jev-1.13 is a commercial endpoint and may change without notice; our results are a snapshot, and we report the exact call count and cost in Appendix R so the snapshot is at least auditable.

Ethics Statement

The corpus is a published, expert-annotated resource of idioms (Sakhawat et al., 2026); it contains no personal data and no content about identifiable individuals. Our interventions rewrite option lists and perturb idiom strings, and produce no new annotation and no generated text.

The work audits two openly released models and one commercial API. We report failure modes in all three, including the vendor’s. We have taken care to separate what is measured from what is inferred. An earlier version of this work reported the commercial arm’s low literal attraction without running it in Bangla; completing the matrix showed that result was a property of the test language, and §5.1 now reports the corrected picture

rather than the flattering one. §5.1 also states the pool selection effect against our own cleanest number. Commercial endpoints are mutable; readers should treat the Jev-1.13 results as dated rather than as a standing property of the product.

Bangla is under-served relative to its speaker population, and we think an audit that locates a failure in the language rather than in the task is the more useful kind to publish. The instrument, the per-item outputs and the analysis code are released; Appendix R is the run record they correspond to.

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Arm	Instruction string
Laya-ml	Bangla, see note
Lod-lille	Bangla, identical to above
Jev-1.13	“What is the actual figurative meaning of this idiom?”

Table 6: The instruction is one sentence and is byte-identical across families, experiments and the two Bangla arms. The Bangla string is the direct equivalent of the English one.

Two confidence signals, and which one we use.

Note that confidence is not the maximum probability: in the example above it is 0.75 while $\max_o p(o) = 0.84$. It is a separate head whose documented job is to estimate whether the correct answer is present in the option list at all. Lod-lille ships the same arrangement. We log it as `conf_head` and report it where it matters (§5.2), but every cross-model number in this paper uses the maximum probability, because it is the only confidence signal all three arms define identically. Where the two disagree the head is not the more diagnostic of the pair: on Lod-lille’s captured items mean head confidence is 0.631 against 0.295 on correct ones, the same inversion the top probability shows.

Answer types we did not use. The interface also accepts `score` (a bounded numeric judgement) and `boolean` questions. We use `choice` exclusively and deliberately. A `score` question would ask the model to rate a gloss in isolation, which destroys the property the instrument depends on: that the gold and the literal reading compete inside a single forward pass, exactly as the psycholinguistic account describes (Cacciari and Tabossi, 1988; Titone and Connine, 1999). A `boolean` question would do the same and additionally reintroduce a yes/no label whose name carries meaning of its own, which is the failure Sun et al. (2026) document. Keeping every experiment in one answer type also means the five distractor families differ in nothing but their option text.

B Positioning against concurrent work

The decision-model literature is a few weeks old at the time of writing, and several papers describe failures adjacent to ours. This appendix states what each one establishes, what it does not, and why our design is complementary rather than duplicative.

Mechanism versus behaviour. Oh et al. (2026) apply causal tracing to autoregressive transformers

and recover two pathways carrying an idiom’s figurative and literal readings at once: early layers and specific attention heads promote the figurative interpretation while a direct route keeps the literal one available, and disambiguating context is integrated from the first layer onward. That is mechanistic evidence that the competition our D-LIT condition exploits is real inside a model, and it is what makes our outcome measure interpretable rather than arbitrary: when a decision model places its mass on the literal gloss, there is a documented sense in which both readings were present and one won.

The two designs answer different questions, and the difference is the reason both are needed. Causal tracing requires weight access, token-level activations and a generative forward pass to intervene on. None of the three systems we audit offers the first, two of them emit no token stream to trace, and one is a commercial endpoint behind an HTTP interface. Their method cannot be pointed at the systems that are currently being deployed as routers and guardrails; ours can, at the cost of saying nothing about where inside the model the competition is resolved. They establish *how* the literal route survives; we measure *how often* it takes over, in which language, and with what confidence attached.

Option names and option definitions. Sun et al. (2026) hold a typed model’s option definitions fixed and permute which name is bound to which, and find decision-flip rates up to 70.4 points higher for semantically loaded labels than for neutral ones: the model follows the name rather than the definition it was given. They note that type-safety conceals this, because every output remains structurally valid.

Our o-series (§5.5) manipulates the complementary half. We hold the option names fixed at A–D and destroy the internal structure of the text bound to them, by truncation and by shuffling words within each option, and find accuracy almost unchanged. Read together, the two results bracket the same conclusion from either side: within an option, these models are sensitive to lexical identity and insensitive to composed meaning. Neither result alone would establish that, because theirs is consistent with a model that reads definitions carefully but weights names more heavily, and ours is consistent with a model that reads definitions as unordered bags.

Context that flips decisions. Xu (2026) append short, fluent, task-preserving context and redirect 61.4% of initially correct decisions, with targeted flip rates of 64.9–73.2% across systems. Our E3 carrier-sentence condition (§5.6) is the benign case of the same mechanism: embedding an idiom in its gold example sentence *raises* accuracy by 0.080 and 0.117 on the two open arms. The two findings are the same sensitivity with opposite signs, and the uncomfortable implication is that a caller cannot tell them apart from the interface. A model whose answer improves when helpful context is appended is, necessarily, a model whose answer moves when context is appended at all.

Calibration and cascades. Ibrahim and Zaki (2026) evaluate decision models against 19 language models on 18 computational social science tasks, and report a median deficit of 11.6 F1 at 44 times lower cost, with confidence calibration *exceeding* most baselines; they recommend decision models as a cheap first stage that routes uncertain items onward. Li et al. (2026) build that system, accepting high-confidence verdicts and escalating the rest, and exceed a frontier model’s accuracy at 41% of its cost.

We reproduce the favourable calibration for the commercial arm on its own language (§5.2), so this is not a contradiction. The boundary we add has two parts. First, in-distribution calibration does not survive a targeted distractor shift, and it fails in the direction that costs a cascade most: on the literal-bearing families the high-confidence tail is where the open arms are *least* accurate, so a confidence gate forwards exactly the items that should have been escalated. Second, and independent of any distribution assumption, the semantics-free control (§5.3) shows what each arm reports when the option list provably carries no information. A gate calibrated on real options has no defence against that case, and a degenerate option list is an ordinary consequence of an upstream bug or an empty retrieval.

What none of this settles. All of the above, ours included, is zero-shot evaluation of base checkpoints. Tang and Zheng (2026) review 28 papers published within nine days of the first widely used release and conclude that the typed readout has shown no independent accuracy advantage, with the demonstrated gains in latency, cost and cascade placement. Whether any of these failure modes survives task-specific fine-tuning is, as far as we can

determine, unmeasured by anyone so far.

C What each experiment changes

Table 8 states, for each experiment, the one thing that varies and the things held fixed. Reading down the “varies” column follows the argument: each row isolates one candidate explanation for the failure measured in E1.

E1, the ladder. D-RAND samples three other idioms’ figurative glosses uniformly. D-SURF samples glosses sharing at least one surface token with the gold. D-SEM samples glosses from idioms sharing at least one semantic tag. D-LIT replaces one sampled distractor with the idiom’s own expert literal gloss. D-HARD applies all three pressures at once. Duplicates are removed and the slate is topped up from the random pool if fewer than k distinct options survive.

E3, perturbing the unit. E3 runs on the literal-bearing option set, so its accuracies are comparable to D-LIT rather than to D-RAND: on the same 1,778 items Laya-ml scores 0.176 under D-LIT and 0.353 under D-RAND, against 0.158 for the intact idiom here. v0 is the intact idiom. v1 shuffles its words, v2 reverses them, v3 deletes one word at random, v4 substitutes one word with a word sampled from another idiom, and v5 replaces the idiom with its gold carrier sentence. The option list is held byte-identical across all six, so the only difference reaching the model is the state string.

E6, ablating the options. o0 is the full gloss. o1 truncates every option to its first six words. o2 shuffles the words within each option. o3 replaces each gloss with its counterpart in the other language. o4 replaces every option with a bare label carrying no semantics, which pins accuracy at chance by construction and is the control of §5.3.

D Why interventions on the option set

This appendix explains the design choice the rest of the paper rests on, for readers who want the reasoning rather than the result.

A decision model takes a question and a list of candidate options and returns a probability over exactly those options. It is not autoregressive, it does not sample, and it emits no text. Three of the usual explainability routes are therefore closed. There is no rationale to inspect and no chain of thought whose faithfulness could be measured (Jacovi and Goldberg, 2020; Turpin et al., 2023). Weights are

option text, as supplied		role	Laya-ml p	Jev-1.13 p
A	একগর্তা; এক সন্তানের জননী; যে নারী একবার মাত্র গর্ভ ধারণ করে	distractor	0.00	0.02
B	ওজন (<i>physical weight</i>)	literal	0.99	0.15
C	ক্ষমতা, গুরুত্ব, মর্যাদা	gold	0.00	0.82
D	হালকাভাবে পড়া	distractor	0.01	0.01

Table 7: **One item exactly as received by the models.** The state is the Bangla idiom ওজন (*ojon*), literally “physical weight,” with the English equivalent *carry weight*. The instruction is identical across items and distractor families. The four option texts are the actual strings reconstructed with the builder and seed used for the run and checked against the recorded gold and literal keys. The probabilities are the returned values. Laya-ml places 0.99 on the literal option, which denotes the physical weight of an object, and 0.00 on the gold figurative gloss; Jev-1.13 places 0.82 on the gold option.

	Varies	Fixed	Seed
E1	distractors	idiom, instr., k	1000
E2	idiom surface	meaning, options	2000
E3	idiom form	options, instr.	3000
E4	task language	item, family	4000
E5	nothing (derived)	n/a	n/a
E6	option text	idiom, gold	6000
E7	option order	option set	7000

Table 8: The seven interventions. E5 issues no calls of its own; it is computed from the E1 outputs. Seeds are the Bangla arms’; the commercial arm uses a disjoint set (Appendix R) because its blocks were run in a different order.

available for two of our arms but not for the commercial one, so no method that requires opening every model, including probing and causal tracing, can be applied uniformly across the three (Oh et al., 2026). And input attribution, the usual fallback (Ribeiro et al., 2016; Lundberg and Lee, 2017), is not only unreliable here (Jain and Wallace, 2019; Adebayo et al., 2018; Hooker et al., 2019) but hard to interpret for this material.

The last point needs stating carefully, because it is what motivates everything else, and the standard shorthand for it is too strong. Idioms are not uniformly non-compositional. Nunberg et al. (1994) argue the opposite of the slogan usually attributed to them: idioms vary in how far their parts can be mapped onto components of the figurative reading, and many are *idiomatically decomposable*, with *spill the beans* distributing its meaning across *spill* and *beans* in a way that *kick the bucket* does not. Psycholinguistic work finds decomposability measurable and consequential for processing (Gibbs et al., 1989; Titone and Connine, 1999).

What follows is weaker than ill-posedness but enough for our purposes. A Shapley value over the tokens of an idiom is always computable, and for a non-decomposable expression there is no reason to expect the resulting per-token shares to correspond to anything linguistically real: the quantity is de-

finied by the attribution game rather than by a decomposition the expression actually supports, and decomposability is a property of the individual idiom that we do not have annotated. The numbers will still arrive, and they will still read as an explanation. We therefore do not build on them, and intervene on the interface instead, which requires no assumption about where an idiom’s meaning sits.

What remains is the interface itself. The caller controls the question and the option list and observes a probability vector. So we intervene there. The corpus supplies, for every idiom, both a gold figurative gloss and an expert literal gloss, which means a counterfactual distractor is already annotated and we create no new labels: dropping the literal gloss into the option list turns a benchmark item into a targeted probe of whether the expression was read as an idiom or as its words. Holding the item, the instruction and k fixed while varying only the distractor family makes the comparison within-item and within-model, so a difference between two families cannot be explained by item difficulty, by the prompt, or by anything about the model that does not change between the two calls.

The same logic produces the other six interventions. If the failure were about the idiom’s surface form rather than its meaning, E2 would show it. If the model held the expression as a stored unit, breaking the unit in E3 would hurt. If the problem were figurativeness rather than the language, E4 would not move. If the model read options as sentences rather than as bags of words, E6 would collapse under scrambling. If the scorer were order-independent as documented, E7 would return the same gloss under all 24 permutations. Each experiment is an attempt to explain the E1 result away, and the paper reports which attempts succeed.

Family	Laya-ml bn	Lod-lille bn	Jev-1.13 en
D-RAND	0.367	0.484	0.968
D-SURF	0.338	0.412	0.915
D-SEM	0.333	0.446	0.829
D-LIT	0.151	0.066	0.931
D-HARD	0.144	0.063	0.854
LAR D-LIT	0.644	0.885	0.047
LAR D-HARD	0.630	0.855	0.043

Table 9: Accuracy by distractor family, and literal attraction on the two families that contain a literal gloss. Chance is 0.250. The commercial column is English and is the reason §5.1 runs the matrix: compare it with the Bangla column of Table 15, where the same model is captured at 0.304.

Coverage	D-RAND	D-LIT	D-RAND	D-LIT
0.05	0.669	0.007	0.977	0.000
0.10	0.597	0.007	0.989	0.000
0.20	0.528	0.023	0.899	0.000
0.35	0.470	0.047	0.785	0.000
full	0.367	0.151	0.484	0.066

Table 10: **Accuracy under a confidence gate.** Cyan identifies Laya-ml, while magenta identifies Lod-lille; the corresponding cell tints are used consistently to distinguish the two model arms. On random distractors, both arms behave as a caller would hope: restricting to the most confident 5% roughly doubles accuracy. On the literal-bearing condition, the curve inverts. Laya-ml falls from 0.151 at full coverage to 0.007 in its most confident 5%, while Lod-lille answers **every one** of its most confident 35% incorrectly. Gating on confidence is therefore worse than not gating, and worse than gating at random.

E The full distractor ladder

Table 9 gives every family for every arm. Two caveats on the commercial column: its numbers are English, and its D-SURF and D-SEM rows run on the larger 8,761-item English pool rather than the 3,265-item literal pool, because those families need no literal gloss.

F Selective prediction in detail

Selective prediction asks what happens if the caller answers only the items the model is most confident about. A useful confidence signal makes accuracy rise as coverage falls. Table 10 reports accuracy at low coverage for the two open arms, on the random-distractor condition and on the literal-bearing one.

This is the result with the most direct operational consequence. A cascade that forwards high-confidence answers and escalates the rest (Li et al.,

2026; Ibrahim and Zaki, 2026) forwards precisely the items on which these models are wrong, and escalates the ones they would have got right.

G Stratified results

Literal capture is not concentrated in rare or obscure idioms, which would be the comfortable explanation. Splitting Laya-ml’s D-LIT condition by the corpus’s own frequency annotation gives LAR 0.671 on *very common* idioms ($n = 1,146$), 0.646 on *common* ($n = 6,200$), 0.615 on *rare* ($n = 1,507$) and 0.565 on *very rare* ($n = 23$). If anything the effect is strongest on the most familiar expressions, which is the opposite of a long-tail story and consistent with these being the idioms whose literal readings are most available.

The corpus’s `cultural_significance` field is unusable as a stratifier: 10,337 of 10,359 rows carry the same value.

H Output distributions

The three arms do not merely differ in accuracy; they place their probability mass very differently, and the shape of that mass is what makes the confidence results of §5.2 possible. Table 12 summarises the returned vectors.

That last point is worth stating separately, because it is the mechanism behind the inverted reliability curves. A model that found the literal option merely distracting would spread mass and lose accuracy together. Both open arms do the opposite: adding the literal gloss *concentrates* the distribution onto it. On 24 items Laya-ml returns $p = 1.000$ on the literal gloss and $p = 0.000$ on the gold.

Worked examples. Table 13 gives four of them. The idioms are transliterated because the toolchain cannot set Bengali script; the released data carries the original strings.

I The capture matrix

Table 15 gives the six cells of §5.1 with the accuracy column alongside LAR. All cells are $n = 3,265$ and were built by one function from seed 21000; 95% bootstrap intervals on LAR are [.716, .746], [.363, .397], [.909, .928], [.227, .256], [.288, .319] and [.040, .055] reading down the table. The paired within-model language effect on capture is +0.352 for Laya-ml, +0.677 for Lod-lille and +0.256 for Jev-1.13, all McNemar

manipulation	string sent to the model	$p(\text{gold})$
E3: the state changes <i>idiom: অগাধ জলের মাছ (Still waters run deep)</i>		
<i>intact</i>	অগাধ জলের মাছ	0.21
<i>word shuffle</i>	মাছ জলের অগাধ	0.19
<i>reversed</i>	মাছ জলের অগাধ	0.19
<i>delete one word</i>	অগাধ জলের	0.24
<i>substitute one word</i>	অগাধ জলের ফ্যা	0.24
<i>in its carrier sentence</i>	সে অগাধ জলের মাছ, কখনো বুঝতে পারি না তার পরিকল্পনা কী।	0.07
E6: the options change <i>gold option rewritten; the other three are rewritten identically</i>		
<i>full gloss</i>	খুব চালাক লোক; সুচতুর ব্যক্তি; যার মনোভাব বোঝা যায় না	
<i>truncated to 6 words</i>	খুব চালাক লোক; সুচতুর ব্যক্তি; যার	
<i>words scrambled</i>	যার সুচতুর চালাক ব্যক্তি; বোঝা খুব না মনোভাব লোক; যার	
<i>English gloss</i>	A very clever person; a shrewd individual; someone whose mindset or intentions are hard to fathom	
<i>semantics removed</i>	A	

Table 11: **What the interventions do to real Bangla text.** The table shows the exact strings used for the idiom অগাধ জলের মাছ (*still waters run deep*, literally “a fish of deep water”). *E3* changes the state supplied to the model; the reported probabilities are those assigned to the gold figurative gloss. Shuffling and reversing the non-compositional expression leave the probability essentially unchanged, whereas the carrier sentence changes it. *E6* changes the option text while holding its role fixed; the five rows show increasingly reduced or rearranged lexical content, ending with the semantics-free control. The other three options undergo the same transformation as the gold option.

		Mean	Med.	>.95	Ent.
Laya	D-RAND	0.438	0.408	0.001	0.892
	D-LIT	0.536	0.479	0.039	0.778
Lod	D-RAND	0.447	0.391	0.002	0.881
	D-LIT	0.745	0.819	0.171	0.497
Jev	D-RAND	0.967	1.000	0.852	0.068
	D-LIT	0.931	0.990	0.708	0.131

Table 12: Distribution of the returned maximum probability, and normalised entropy of the full vector (1 = uniform). The literal-bearing family makes both open arms *sharper*, not flatter: Lod-lille’s mean peak rises from 0.447 to 0.745 and the share of near-deterministic answers from 0.002 to 0.171, while its accuracy falls from 0.484 to 0.066. The model is not confused by the literal gloss; it is convinced by it.

Idiom	Literal	Figurative (gold)	p_{lit}
tel	oil	arrogance	1.000
chamcha	spoon	sycophant	1.000
til	sesame seed	a tiny amount	1.000
charpaya	four-legged	rope cot	1.000

Table 13: Laya-ml under D-LIT. In each case the expert literal gloss and the expert figurative gloss are both in the option list, and the model places all of its mass on the literal reading. These are not hard or obscure idioms; three of the four are single words in everyday use.

$p < 10^{-100}$. Jev-1.13’s English cell reproduces the value reported from the original run, 0.0472, which is why we treat the two constructions as mea-

suring the same thing.

J How the two language cells were built

The capture matrix is the result we lean on hardest, so this appendix states exactly how its two language cells were constructed, lists every asymmetry between them that we could find, and reports the one test of the largest asymmetry that the existing outputs permit.

Where the glosses come from. Neither literal gloss is a translation produced by us or by a machine. The corpus annotates one field per idiom, *literal_meaning*, written in Bangla with an English parenthetical alongside it, so a single expert-authored field carries both readings. The Bangla cell uses that field as written. The English cell uses the parenthetical alone, recovered by a regular expression that accepts a parenthesised span only if it contains at least three ASCII letters and is at least 60% ASCII by character, which excludes Bangla asides and bare numerals. Of 8,876 items, 3,391 carry such a parenthetical, 3,352 of those also carry an English equivalent idiom, and 3,265 survive a final cut requiring the English literal gloss to be lexically distinct from the English gold gloss (Jaccard < 0.34). That pool of 3,265 is the matrix. The provenance matters because a machine-

idiom	expert literal gloss	gold figurative gloss	English equivalent	$p(\text{literal})$
ঈশ্বরপ্রাপ্তি	ঈশ্বরকে অধিগ্রহণ (attainment of God)	মৃত্যু (death)	kick the bucket	0.97
আকাশের চাঁদ	আকাশের চাঁদ (sky's moon)	অজস্র বা দুর্লভ বস্তু; বহু কল্পিত বস্তু (an unattainable or rare object; an imagined object)	pie in the sky	0.99
অল্প জলের মাছ	অল্প জলের মাছ (little water's fish)	সামান্য পুঁজির বা সম্পদের মালিক; অল্প ধন নিয়ন্ত্রিত ব্যক্তি (a person with limited capital or resources; a poor person with little wealth)	big frog in a small puddle	0.97
ওজন	ওজন (physical weight)	ক্ষমতা, গুরুত্ব, মর্যাদা (power, importance, status)	carry weight	0.99
অগ্নিপরীক্ষা	আগুনের পরীক্ষা (fire test)	কঠিন পরীক্ষা; কঠিন পরিস্থিতির সম্মুখীন (a difficult test; facing a difficult situation)	trial by fire	0.96

Table 14: **Items where Laya-ml places almost all probability on the literal gloss.** Each item contains the Bangla idiom, the expert-provided literal gloss, the gold figurative gloss, and the English equivalent recorded by the annotators. English translations in parentheses are provided for readability; the underlying Bangla strings are the original dataset content. Both literal and figurative glosses were present in the option list. The rightmost column reports the probability assigned to the literal gloss. These are not obscure expressions: *ishwarprapti* corresponds to *kick the bucket* and is read literally as “attainment of God,” while *akashar chand* corresponds to *pie in the sky* and is read literally as “the moon of the sky.”

	Bangla		English	
	acc	LAR	acc	LAR
Laya-ml	0.127	0.731	0.417	0.379
Lod-lille	0.044	0.918	0.684	0.241
Jev-1.13	0.536	0.304	0.926	0.047

Table 15: The literal condition as a model $\times$ language matrix, all six cells measured on the same 3,265 items with one construction and one seed ($n = 3,265$ per cell; 95% CIs in Appendix I). Bold marks the highest LAR per language. Every arm is far more captured in Bangla than in English, and the Bangla ordering does not follow capability: Jev-1.13 is the most accurate and the least captured, while Lod-lille is more accurate than Laya-ml and more captured.

translated English gloss would make the English cell a measurement of the translator; an annotator-written one does not.

The four asymmetries. We can state these precisely, which is better than asserting the cells are matched.

(i) *The state differs in kind.* The Bangla state is the idiom itself. The English state is the annotator’s nearest English idiom, taken from the `similar_in_english` field, not a translation of the Bangla one. The cells are therefore paired entries, and we use that phrase in the paper rather than “identical items”.

(ii) *The Bangla literal option is bilingual.* Because the Bangla cell uses the annotated field as written, and pool membership requires that field

to contain an English parenthetical, every one of the 3,265 Bangla literal options contains its own English gloss inside it. It averages 44 characters against 24 for the English literal option. We did not strip the parenthetical, because doing so would have meant presenting the open arms a string the corpus does not contain; the cost is that the Bangla literal distractor is longer and partly in the other script.

(iii) *The distinctness cut applies to one side only.* The Jaccard < 0.34 filter between gold and literal gloss was applied in English and not in Bangla. Measured after the fact, the filter turns out not to bind: mean gold-versus-literal Jaccard is 0.029 in Bangla against 0.052 in English, so the Bangla side is the more separated of the two and would have passed the cut almost everywhere. This asymmetry is procedural rather than consequential.

(iv) *The literal gloss echoes the state more often in Bangla.* This is the one that could plausibly manufacture the result. The Bangla literal gloss shares at least one content word with the Bangla idiom on 56.2% of pool items; the English literal gloss shares one with the English equivalent idiom on 25.1%. A model doing order-free lexical matching on the option side, which §5.5 shows all three partly are, has a surface hook into the Bangla literal option that it lacks in English.

Testing asymmetry (iv). The existing per-item outputs are enough to test it without running any-

	Laya	Lod	Jev
<i>Bangla LAR by surface hook</i>			
hook absent ($n = 1429$)	0.716	0.845	0.376
hook present ($n = 1836$)	0.742	0.975	0.247
difference	+0.026	+0.130	-0.129
<i>Capture gap, hook-absent items only</i>			
Bangla LAR	0.716	0.845	0.376
English LAR	0.358	0.236	0.066
gap	+0.358	+0.610	+0.310
headline gap (all pool)	+0.352	+0.677	+0.256

Table 16: **The construction asymmetry does not carry the result.** Laya-ml, Lod-lille, Jev-1.13. “Hook” means the Bangla literal gloss shares at least one content word with the idiom, which is the asymmetry most likely to inflate Bangla capture. Upper block: capture is not reliably higher when the hook is present, and on the commercial arm it is lower. Lower block: restricting both language cells to hook-absent items leaves the capture gap essentially unchanged, and larger for two of three arms. The script is released with the per-item outputs.

thing new. Split the pool on whether the Bangla literal gloss shares a content word with the idiom and read LAR in each stratum (Table 16, upper block). If the hook were driving capture, the overlap stratum would be far more captured in every arm. It is not: the effect is +0.026 for Laya-ml, +0.130 for Lod-lille and -0.129 for Jev-1.13. The sign is not even consistent across arms, and on the commercial arm the hook is associated with *less* capture, which is the opposite of the predicted direction.

The sharper test restricts both language cells to the 1,429 items where the hook is absent and re-reads the gap there (lower block). It barely moves: +0.358, +0.610 and +0.310, against headline gaps of +0.352, +0.677 and +0.256. For two of the three arms the gap is slightly *larger* once the hook is removed. We read this as showing that asymmetry (iv) does not account for the Bangla–English difference, not as showing the cells are equivalent; a corpus paired by translation rather than by annotator judgement would settle what we can only bound here.

What this appendix does not establish. It tests one asymmetry, the one we could operationalise from recorded outputs. Asymmetries (i) and (ii) are structural and cannot be tested without new data: we cannot ask what the arms would do on a Bangla literal option with its parenthetical stripped, or on an English state that is a literal translation rather than an idiomatic equivalent, without run-

Family	Full	Subset	Drift
D-RAND	0.367	0.351	-0.016
D-SURF	0.338	0.339	+0.001
D-SEM	0.333	0.325	-0.007
D-LIT	0.151	0.136	-0.015
D-HARD	0.144	0.149	+0.004

Table 17: Laya-ml on the 10% subset against its own full set. Maximum drift is 0.016. Two of five subset estimates fall marginally outside the full-set bootstrap CI, which is expected at $n = 888$ and changes no sign or ordering. The full-set figures are therefore reported as the precise estimates.

ning those conditions. We flag them in Limitations rather than dismissing them.

K Cross-lingual arms

Figure 7 plots the E4 conditions discussed in §5.7.

L The matched three-way condition

Figure 8 plots Table 3, the one condition that is identical across all three arms.

M Unit integrity

Figure 9 plots the E3 perturbations described in §5.6. The options are byte-identical across all six variants, so the only difference reaching the model is the state string.

N Surface invariance

The corpus ships alternative surface forms certified by annotators as meaning-preserving. A model reading meaning should be invariant across such a pair. Laya-ml agrees with itself on 0.657 of 3,143 pairs (2,423 items) and Lod-lille on 0.677 of 310 pairs (234 items): about a third of the time, a meaning-preserving rewrite changes the answer. Mean Jensen–Shannon divergence between the two probability vectors is 0.039 and 0.085.

O Subset representativeness

Lod-lille ran a fixed 10% random subset. Table 17 compares Laya-ml’s accuracy on those same 888 items against its own full 8,876.

P Full option ablation

Table 18 gives the numbers behind Figure 10.

Q Order sensitivity and position bias

This appendix reports the per-slot selection shares behind §5.4, then sets out the permutation au-

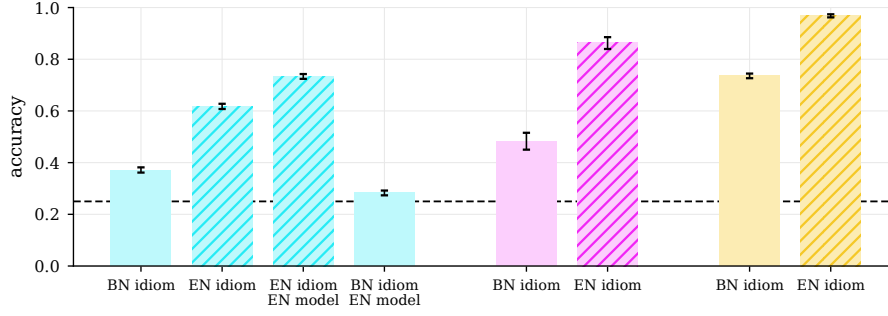


Figure 7: Accuracy by task language, grouped by model: Laya-ml (four bars), Lod-lille, Jev-1.13. Solid bars are Bangla input, hatched bars English; Laya-ml’s third and fourth bars use the English-only checkpoint, the fourth being the control. Error bars 95% bootstrap CIs, dashed line chance. Every model gains 20–45 points from the same idiom presented in English.

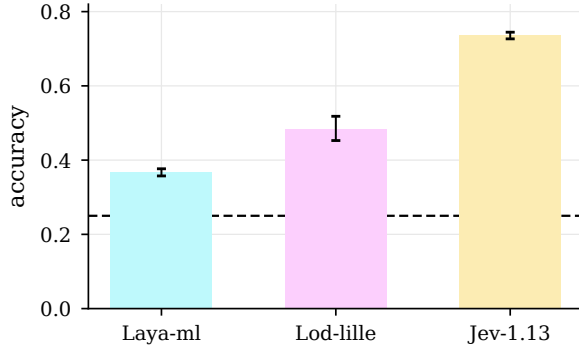


Figure 8: The matched three-way comparison. Bars left to right: Laya-ml, Lod-lille, Jev-1.13. Black dashed line is chance, error bars 95% bootstrap CIs. This capability ordering is the baseline against which the calibration failures in §5.2–§5.3 should be read.

Option text	Laya	Lod	Jev
full gloss	0.365	0.492	0.970
truncated	0.361	0.468	0.958
scrambled	0.357	0.479	0.974
other lang.	0.347	0.398	0.963
no semantics	0.256	0.233	0.254

Table 18: **Option-side ablation.** The semantics-free row removes meaningful option content and yields near-chance accuracy in all three arms. Jev-1.13’s column is English.

dit that establishes the residual instability of the independent-option scorer is real rather than an artifact of our harness.

What is held fixed. Each item here is one idiom with one fixed set of four option texts. We present that same set 24 times, once per permutation, and change nothing else: the instruction string, the state, k , the decoding configuration and the option texts themselves are byte-identical across the 24 calls, and no call is retried. The only quantity that varies is which slot each option occupies. Anything that moves across the 24 calls is therefore attributable to order, which is what makes this the

cleanest of our seven interventions.

Recovering option identity. The harness records, per call, the permutation as a four-digit string in which position s holds the index, in the canonical option list, of the option displayed in slot s . A chosen slot must be mapped back through that permutation before it can be compared across calls. Counting distinct slot *letters* instead inflates instability to near unity for any scorer, because a letter necessarily changes whenever the same gloss is moved, and we report this because an earlier draft of our own analysis made that mistake. All numbers in the paper use the mapped gloss identity.

The permutation audit. Table 19 reports a quantity that the summary measures in Table 5 cannot: not whether the *answer* changes, but how far a *single option’s own score* travels as that option is moved around the list. This is the quantity the independent-option design makes a claim about. Scoring each option in its own forward pass should make an option’s score a function of the option and the state alone, so the spread should sit at the level of floating-point accumulation noise, which for fp32 summation over sequences of this length is on the order of 10^{-7} . It does not. Lod-lille’s median item has an option whose score moves 4.0×10^{-2} , five orders of magnitude higher, and not one of its 299 items falls below 10^{-4} . Two consequences follow. First, the 10.7% answer-change rate in Table 5 is not a tie-breaking artifact: ties would require near-identical scores, and the scores are not near-identical. Second, the documented guarantee is not one an evaluation can rely on, at least in the configuration we were able to run.

The commercial arm. Jev-1.13 makes no public invariance claim, and its behaviour is the least

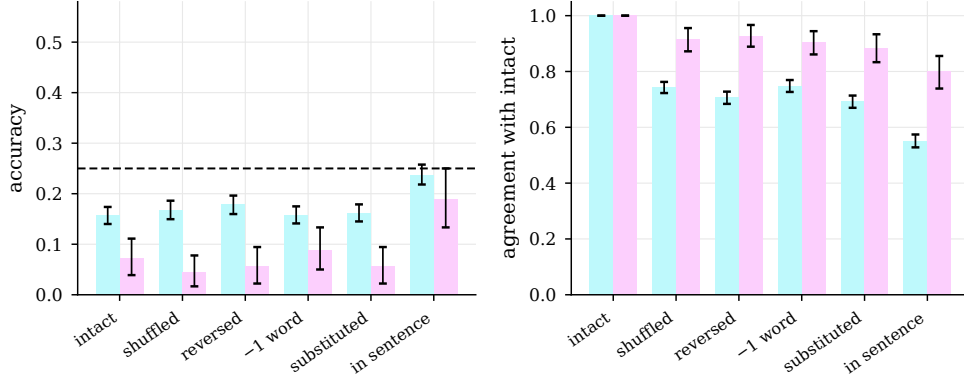


Figure 9: Unit integrity: the idiom is perturbed, the options held fixed. ■ Laya-ml, ■ Lod-lille; error bars 95% bootstrap CIs, dashed line chance (left panel). *Left*: accuracy. *Right*: agreement with the answer on the intact idiom. Scrambling and reversing a non-compositional expression does not meaningfully hurt either model, which is the bag-of-words signature.

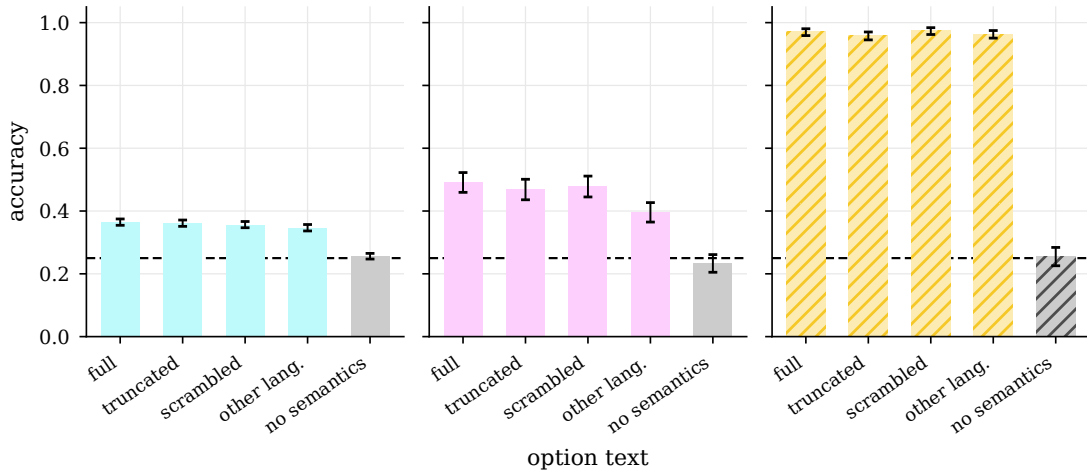


Figure 10: Option-side ablation. Panels left to right: ■ Laya-ml, ■ Lod-lille, ■ Jev-1.13. In each panel the fifth, ■ bar is the semantics-free control of §5.3; the whole Jev-1.13 panel is hatched because its task is in **English**. Error bars are 95% bootstrap CIs and the black dashed line is chance. Truncating and word-scrambling the option text cost almost nothing in any of the three models.

like either open arm: it is the only arm that returns bit-identical distributions on some items at all, on 23.0% of them, yet its mean spread (0.139) is three times Lod-lille’s and its maximum (0.820) approaches Laya-ml’s. The distribution is bimodal rather than merely noisier, which is what one would expect from a scorer that is close to deterministic on easy items and switches regime on hard ones. We cannot test that reading without weights, and we do not claim it.

Position bias. Figure 11 gives the per-slot selection shares. Because the gold option is placed uniformly at random, the expectation under no positional bias is exactly 0.25 in every slot for every arm, so this figure needs no baseline model: the distance from the dashed line *is* the bias. Laya-ml’s profile is not the primacy gradient usually reported for prompted generative scorers (Zheng et al., 2024); it peaks on slot B at 37.2% against

14.6% for slot A, which is a $2.6\times$ accuracy swing from position alone and is not a monotone function of position at all. We note the shape and decline to explain it; with one model per scoring design, we have no way to separate a property of the marker scheme from a property of this particular backbone.

R Run record

Environment. Python 3.13.15 throughout. The two open arms ran on one Tesla T4 (15,360 MiB): Laya-ml in bf16 with max_len 1024 at 27.2 ms per call, Lod-lille in fp32 at 86.1 ms per call. The commercial arm needs no accelerator and ran on CPU against typesafe/jev-1.13.

Temperatures. Laya-ml’s run recorded temperature = [1.0, 1.0, 1.0] with temperature_by_options empty, so the per-bucket temperatures it ships were not applied

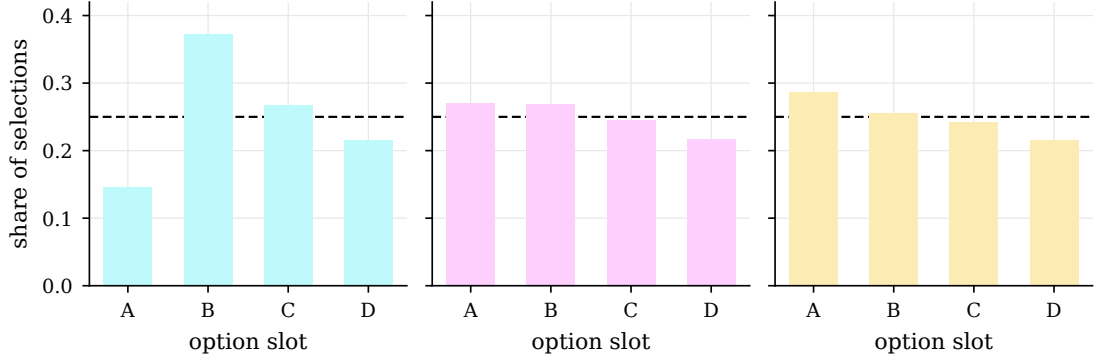


Figure 11: Share of selections falling on each option slot. Panels left to right: Laya-ml, Lod-lille, Jev-1.13. The gold option is placed uniformly at random, so the black dashed line at 0.25 is the expectation under no positional bias and any departure from it is bias. Laya-ml peaks on slot B, taking it 37.2% of the time against 14.6% for slot A; the other two arms sit close to the line throughout, with a mild and monotone preference for earlier slots.

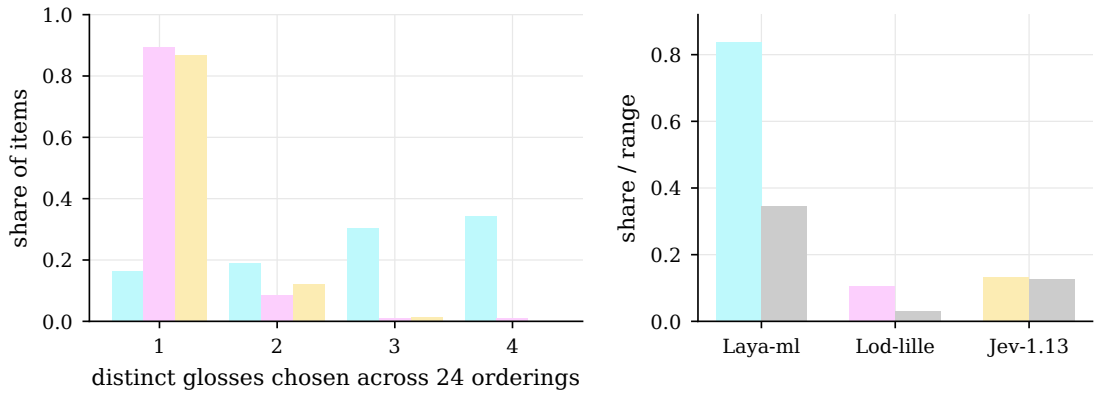


Figure 12: Order sensitivity. *Left*: how many distinct glosses each model selects across the 24 orderings of one option set; within each group bars run Laya-ml, Lod-lille, Jev-1.13, and a value of 1 means perfect stability. *Right*: per model, the coloured bar is the share of items whose chosen gloss is unstable and the grey bar is the mean within-item range of $p(\text{gold})$. Laya-ml is unstable on 83.7% of items with a mean $p(\text{gold})$ swing of 0.346; the other two are near-stable.

	Laya	Lod	Jev
items	992	299	300
median spread	0.495	0.040	0.070
mean spread	0.507	0.046	0.139
95th pct. spread	0.772	0.105	0.480
max spread	0.928	0.245	0.820
share $< 10^{-4}$	0.000	0.000	0.230
answer changes	0.837	0.107	0.133

Table 19: **Permutation audit.** Per item, the largest distance a single option’s own score travels across the 24 orderings of its list, after mapping each slot back to the option it displayed. Laya-ml, Lod-lille, Jev-1.13. Bold marks the largest value in each row. An order-independent scorer should sit at fp32 noise, around 10^{-7} ; the *share* row gives the fraction of items at or below 10^{-4} , a threshold three orders of magnitude more permissive than that. Only Jev-1.13 returns bit-identical distributions on any item, and then on fewer than a quarter of them. The script that produces this table is released with the per-item outputs.

and its ECE figures are raw. Lod-lille ships temperature: 1.0 by design, its authors hav-

ing reported that fitting hurt held-out calibration. Both open arms are therefore untempered and comparable on that axis. The commercial arm’s internal calibration is not observable.

Seeds. The Bangla arms use a per-experiment seed (E1 1000, E2 2000, E3 3000, E4 4000, E6 6000, E7 7000) added to the item id, so each item’s option permutation is reproducible and identical across arms. The split seed is 20261001. The commercial arm uses a disjoint block seed set (order audit 17000, literal 12000, ceiling 11000, Bengali control 14000, options 16000) because its blocks ran in a different order; the permutation for a given item is still a pure function of seed and id.

Call volume and cost. The commercial arm issued 56,533 calls for \$0.9348 at \$0.042 per million input tokens with output free. The blocks are the order audit ($300 \times 24 = 7,200$), literal capture on the shared pool ($3,265 \times 3 = 9,795$), the English ceiling ($8,761 \times 3 = 26,283$), the Bengali control

(8,761) and the option ablation ($873 \times 5 = 4,365$), which is 56,404 retained rows. The remaining 129 calls are the endpoint probe and the response-contract checks made before each block. Transient 429 and 5xx responses were retried with exponential backoff to a 30 s ceiling; no call was dropped.

Statistics. All intervals are 95% percentile bootstrap over items, 2,000 resamples, fixed seed 0. ECE uses 15 equal-width bins. Paired comparisons use McNemar’s exact test via the binomial distribution. AURC is reported against a random-gate baseline computed on the same items.

What is released. The per-item outputs for every cell of every experiment (one row per item per condition, carrying the full probability vector, the predicted key, the gold key, the literal key and the gold position), the five split manifests, the three run environment records quoted above, the task configurations containing the verbatim instruction strings, the analysis scripts that turn those outputs into every table in this paper, the figure sources, and a CSV of the exact plotted values beside each figure.

Pairing. Because the two open arms were reconstructed independently from a written contract rather than from shared code, the released outputs admit a direct check: on the 4,440 overlapping rows the gold key, gold position, idiom string and literal-distractor key agree at 1.000. That check is itself a released artifact, and it is what licenses every paired test in the paper.

The matrix run. The six cells of §5.1 were run afterwards as one job, with its own seed namespace (21000) so no option permutation collides with the runs above. Every cell covers the same 3,265 items and is built by a single function; the two languages share one distractor draw per item and are shuffled independently from the same generator. Laya-ml ran both languages on the multilingual checkpoint, so only the language changes and the checkpoint does not. Lod-lille ran the full pool rather than its earlier 10% subset, which supersedes the thin 316-item in-pool estimate. The commercial arm issued 6,531 calls for \$0.1152. Its English cell reproduces the value from the original run to four decimal places (0.0472), which is the check that the two constructions measure the same thing.

Known gaps. E2 was not run on the commercial arm, because the alternative surface forms exist only in Bangla and the arm’s task is English. The

matrix covers the literal condition only; the other distractor families were not run in both languages. The confidence and calibration results remain two-arm, because the commercial arm’s reliability was measured on its English run.